

AFSAL A AZEEZ

Senior Full-Stack & AI Engineer

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Abu Dhabi, UAE • **Available Immediately**

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Professional Summary

Full-stack and AI engineer with 5+ years of experience building and shipping production-grade software at a Mayfield-funded AI startup. Deep hands-on experience across the entire stack — React/Next.js frontend, Node.js and Java/Spring Boot backend, Docker/Kubernetes infrastructure, and LLM integration. Proven track record of end-to-end ownership: from context engineering and AI pipeline design to enterprise deployments. IIT Bombay M.Tech (GATE AIR-241, top 0.22% nationally); degree fully attested for UAE residency compliance. Immediately available in Abu Dhabi.

Technical Skills

- **Frontend:** React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5/CSS3, Redux Toolkit, Zustand, Tailwind-CSS, Electron.js
- **Backend:** Node.js (TypeScript), Python, Java (Spring Boot), Go, REST/GraphQL APIs, microservices, Web-Sockets
- **AI & LLM:** LLM integration (OpenAI, Gemini, Claude, Llama 3), prompt & context engineering, multi-provider orchestration, AI response caching
- **Infrastructure:** Docker, Kubernetes, AWS (EC2, S3, Lambda, EKS), Terraform, Helm, CloudFormation, GitHub Actions, OpenShift
- **Databases & Testing:** PostgreSQL, SQLite, MongoDB, Redis, Elasticsearch; Playwright, Jest, OWASP ZAP

Professional Experience

Member of Technical Staff

Sep 2020 – May 2026

Full-Stack & AI Engineer

Zettabytes Inc. (Roost.ai) — Mayfield-funded DevOps & AI Startup

Phase B — RoostGPT: AI Test Generation Platform

2023 – 2026

- **Java Unit Test Generation — End-to-End Ownership:** Owned the complete Java unit test generation pipeline — from repository ingestion to final generated test files. Engineered LLM context by working directly on the **Spring Boot Java AST parsing service** to extract class graphs, bean dependency trees, and method signatures from multi-module Maven/Gradle repositories. Achieved a **93% compile success rate** across generated test suites.
- **Test Scenario Analysis & Report UI:** Built end-to-end (frontend, API, and backend) — listing covered/uncovered scenarios, enabling targeted generation triggers, and displaying summaries, compile checks, and coverage output. Delivered **4× improvement in test coverage** and **85% reduction in manual testing time**.
- **Java AST Schema Redesign:** Flattened nested class structures into first-class entities, reducing per-class file size from **316KB to 17KB** (95% reduction) — improving LLM context efficiency and test generation accuracy.
- **Git Platform Integrations:** Built integrations within RoostGPT connecting to GitHub, GitLab, Bitbucket, and Azure Repos for automated repository access, test generation triggers, and direct commit workflows. Implemented PR webhook hooks to retrigger generation on new code changes.
- **Microsoft Teams Notification Bot:** Took end-to-end ownership of the Teams notification bot (Azure Bot Framework), delivering real-time test generation status and CI/CD notifications within Teams channels.
- **Multi-LLM Provider Dashboard & AI Caching:** Built the frontend for switching between LLM providers (OpenAI, Gemini, Claude, Llama 3). Designed an SQLite-based caching layer for LLM responses, eliminating redundant API calls and reducing latency.
- **Frontend & Mentorship:** Built the complete test generation workflow UI — repo analysis views, trigger dashboards, review interfaces, Playwright E2E suites. Mentored junior engineers and interns; conducted code reviews via JIRA.
- **Tech Stack:** React.js, Next.js, TypeScript, Node.js, Spring Boot (Java), SQLite, LLM APIs, Docker, GitHub Actions.

Phase A — Roost Desktop, Serverless & EaaS Platform

2020 – 2022

- **Cross-Platform Desktop App:** Built and maintained an **Electron.js + React.js** desktop application for DevOps collaboration (macOS/Windows/Linux). Developed Kubernetes topology visualisation UI (pod networks, sidecar connections, cluster state).

- **Serverless Migration & EaaS:** Migrated Roost Desktop to a serverless AWS Lambda architecture. Built EaaS features: multi-repo commit, env var injection, Helm chart filtering, CloudFormation and Terraform UI.
- **Enterprise Deployments:** Supported on-premise and OpenShift deployments for enterprise clients across the US market, collaborating directly with their engineering and DevOps teams on cluster integration and platform rollout.
- **Tech Stack:** React.js, Node.js, Python, Go, REST/GraphQL APIs, Kubernetes, Docker, Terraform, AWS (EKS, EC2, S3, Lambda), Electron.js.

Education

Master of Technology (M.Tech.) in Computer Science & Engineering <i>Indian Institute of Technology (IIT), Bombay</i>	Graduated 2020 <i>Mumbai, India</i>
• GATE AIR-241 — Top 0.22% out of 107,893 candidates. Degree fully attested for UAE residency compliance.	
Bachelor of Technology (B.Tech.) in Computer Science & Engineering <i>Mar Athanasius College of Engineering (MACE)</i>	Graduated 2017 <i>Kerala, India</i>

Projects & Teaching

- **VisionLog-MLX:** MLX-VLM powered vision logging system using Qwen2-VL-2B-Instruct for real-time camera feed analysis. (Open-source)
- **GIS Water Network System — IIT Bombay:** GIS-integrated database in QGIS with custom Python plugins for spatial data extraction; EPANET and JalTantra workflow integration.
- **Teaching Assistant — IIT Bombay (2018–2019):** CS699 Software Lab and CS101 Computer Programming; mentored 14 students individually; led lab sections for 400+ undergraduates.